Polyagamma Augmentation in GP Classification using Equispaced Fourier Features

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Contents

1 Background

1.1 GP setup

Assume $(x_i, y_i) \in \mathbb{R}^d \times \{0, 1\}$ observations come from:

$$y_i \mid f, x \sim \mathrm{Bern}\big(\sigma(f(x_i))\big) \quad \text{independently}$$

$$f \sim GP(0, K_\theta),$$

Our goal is to learn hyperparameters θ, then compute the predictive mean and variance.
Throughout we use our equispaced fourier feature basis $K \approx \Phi\Phi^* = FD^2F^*$.

1.2 Polyagamma Augmentation

Introduce auxiliary

$$\omega_i \sim PG(1, 0)$$

. Use two identities:

1. $\frac{(e^u)^y}{(1+e^u)} = \frac{1}{2}\exp((y - \frac{1}{2})u) \int \exp(-\frac{1}{2}\omega u^2) PG(\omega|1, 0) d\omega$
2. $\exp\left[-\frac{1}{2}\,\omega\, c^2\right]\, PG(\omega \mid 1, 0) \propto PG(\omega \mid 1, c)$

$$p(y, f) = p(f) \prod_{i=1}^{n} \frac{1}{2} \exp\left[(y_i - 1/2)\, f(x_i)\right] \int \exp\left[-\tfrac{1}{2}\,\omega_i\, f(x_i)^2\right] PG(\omega_i \mid 1, 0)\, d\omega_i \tag{1}$$

$$\implies p(f, y, \omega) \;\propto\; p(f) \prod_{i=1}^{n} \exp\left[(y_i - 1/2)\, f(x_i)\right] \exp\left[-\tfrac{1}{2}\,\omega_i\, f(x_i)^2\right] PG(\omega_i \mid 1, 0). \tag{2}$$

$$\boxed{\omega_i \mid f, y \sim PG(1, f(x_i)) \text{ independently}},$$

$$\boxed{f \mid \omega, y \sim \mathcal{N}(m, V)},$$

where

$$V = \left(K_\theta^{-1} + \Omega\right)^{-1}, \quad m = V\,(\,y - \tfrac{1}{2}\,).$$

2 E step

Fix Kernel hyperparameters θ; update variational distribution $q(\omega, f)$.

2.1 Warmup; CAVI Updates

Use mean-field variational family; posit:

$$q(\omega, f) = q(f) \prod_{i=1}^{n} q(\omega_i).$$

Optimize q via CAVI:

1. First: **Fix** $q(\omega)$.

$$\begin{aligned}
\arg\max_{q(f)} \mathbf{E}_{q(f,\omega)} \log \frac{p(y, f, \omega)}{q(f, \omega)} &= \arg\max_{q(f)} \mathbf{E}_{q(f)} \mathbf{E}_{q(\omega)} \log \frac{p(f \mid y, \omega)}{q(f)} && (3)\\
&= \arg\min_{q(f)} \mathrm{KL}\big(q(f) \,\|\, p(f \mid y, \mathbf{E}_{q(\omega)}[\omega])\big) && (4)\\
&= \mathcal{N}\big(\Sigma\,(y - \tfrac{1}{2}),\, (K_\theta^{-1} + \Lambda)^{-1}\big), && (5)
\end{aligned}$$

where $\Sigma = (K_\theta^{-1} + \Lambda)^{-1}$. Ie, the solution is the posterior $p(f \mid y, \omega)$ evaluated at $\omega_i = \mathbf{E}_{q(\omega)}[\omega_i]$. [1]

We reach **??** because $\log p(f|y, \omega)$ is linear in ω:

$$\begin{aligned}
\log p(f(x) \mid y, \omega) &= -\tfrac{1}{2}\,(f(x) - \Sigma(y - \tfrac{1}{2}))^T\,\Sigma^{-1}\,(f(x) - \Sigma(y - \tfrac{1}{2})) + \text{const} && (6)\\
&= -\tfrac{1}{2}\, f(x)^T\,(K_\theta^{-1} + \Omega)\, f(x) + \ldots && (7)
\end{aligned}$$

So $\mathbf{E}_{q(f)} \log p(f \mid y, \omega) = p\big(f \mid y, \mathbf{E}_q[\omega]\big)$.

2. **Fix** $q(f)$. Recall that we can write $PG(\omega_i|1, c_i) = \exp(-\frac{1}{2}\omega_i c^2) PG(\omega_i|1, 0)$. This property of PG yields:

$$\arg\min_{q(\omega)} \mathbf{E}_{q(\omega)} \mathbf{E}_{q(f)} \log \frac{PG(\omega_i \mid 1, f(x_i))}{q(\omega_i)} = \arg\min_{q(\omega)} \mathrm{KL}\big(q(\omega) \,\|\, PG(\omega_i \mid 1, c_i)\big), \tag{8}$$

where $c_i = \sqrt{\mathbf{E}_{q(f)} f(x_i)^2} = \sqrt{\mathbf{Var}_{q(f)}[f(x_i)] + \big(\mathbf{E}_{q(f)}[f(x_i)]\big)^2}$ So we have

$$q(\omega_i) = PG(\omega_i \mid 1,\ c_i),$$

2.2 CAVI Summary

$$\boxed{q(f) = \mathcal{N}(\mu, \Sigma)}, \qquad \boxed{q(\omega_i) \sim PG(1, c_i)}$$

where we've defined:

$$c_i = \sqrt{\mathbf{Var}_{q(f)}[f(x_i)] + (\mathbf{E}_{q(f)}[f(x_i)])^2}, \quad \Lambda = \mathrm{diag}\big(\mathbf{E}_q[\omega_i]\big) = \mathrm{diag}\big(\tfrac{\tanh(c_i/2)}{2c_i}\big), \quad \Sigma = (K_\theta^{-1} + \Lambda)^{-1}, \quad \mu = \Sigma(y - \tfrac{1}{2}).$$

2.3 Natural Gradient Updates

Let's assume no inducing points.

Natural[2] parameters:

$$\eta_1 = \Sigma^{-1}\mu, \quad \eta_2 = -\tfrac{1}{2}\,\Sigma^{-1}.$$

Supposing we work with minibatches S with $s = |S|$, Wenzel Eq 12 gives the natural gradient updates:

$$\begin{aligned}
\eta_1^{(t)} &\leftarrow (1 - \rho_t)\,\eta_1^{(t-1)} + \rho_t\,\tfrac{n}{s}\,\kappa_S^T\,(y_S - \tfrac{1}{2}),\\
\eta_2^{(t)} &\leftarrow (1 - \rho_t)\,\eta_2^{(t-1)} - \tfrac{\rho_t}{2}\big[K_\theta^{-1} + \tfrac{n}{s}\,\kappa_S^T \Lambda_S \kappa_S\big].
\end{aligned}$$

[1] Can also show with $\beta \sim N(0, I)$ that we could get $q(\beta) = \mathcal{N}\big(m,\, (I + \Phi^* \Lambda \Phi)^{-1}\big)$.

[2] Algorithm 3 in https://arxiv.org/pdf/1711.06999

If we don't use sparse inducing points method, we get

$$\kappa_S = K_{Sn} K_{nn}^{-1} = I_S \in \mathbb{R}^{s \times n} \quad \text{(row selectors)},$$

yielding

$$\eta_1^{(t)} \leftarrow (1-\rho_t)\, \eta_1^{(t-1)} + \rho_t \, \tfrac{n}{s} \, I_S^T \, (y_S - \tfrac{1}{2}),$$
$$\eta_2^{(t)} \leftarrow (1-\rho_t)\, \eta_2^{(t-1)} - \tfrac{\rho_t}{2} \left[K_\theta^{-1} + \tfrac{n}{s} \, I_S^T \Lambda_S I_S \right].$$

Now consider the η_2 update. Wenzel introduces m inducing points, so that $\eta_2 \in \mathbb{R}^{m \times m}$. However, for non-sparse GP ($m = n$), then storing and updating an $n \times n$ matrix is impractical. But for the M-step, all we have to store is how to apply Σ The proposed solution is to first rewrite:

$$-2\eta_2^{(t)} = \Sigma_{(t)}^{-1} = K_{\theta^{(t)}}^{-1} + \Delta^{(t)} \leftarrow (1-\rho)\big(K_{\theta^{(t-1)}}^{-1} + \Delta^{(t-1)}\big) + \rho\big(K_{\theta^{(t)}}^{-1} + \tfrac{n}{s} I_S^T \Lambda_S I_S\big),$$

Instead of updating Σ, update:

$$\Delta^{(t)} \leftarrow (1-\rho)\, \Delta^{(t-1)} + \rho \, \tfrac{n}{s} \, I_S^T \Lambda_S I_S.$$

then either

1. Store the function:

$$v \mapsto \Sigma_{(t)}^{-1} v = (1-\rho) K_{\theta^{(t-1)}}^{-1} v + \rho (K_{\theta^{(t)}}^{-1} v) + \Delta^{(t)} v,$$

   for use in the M-step. Or:

2. Run enough updates of natural parameters before moving to the M-step that $(1-\rho)^k$ is negligible. Or:

3. Simply "ignore" the old kernel; after M-step, reset $\Sigma^{(t-1)} = K_{\theta^{(t)}}^{-1} + \Delta^{(t-1)}$.

I think the last option is most practical, if it works. We avoid applying this convex mix of K^{-1} operators in the M-step, which would be costly and confusing. Then since θ will be constant within the E-step, we can just update Δ and then store the apply $v \mapsto \Sigma_{(t)}^{-1} v = (K_\theta^{-1} + \Delta^{(t)}) v$.

2.4 E step Summary

Alternate:

1. Calculate c_i using $\eta_2^{(t-1)}, \eta_1^{(t-1)}$.

2. Update η_1, η_2 by storing diagonal Δ and doing batch coordinate ascent,

2.5 E-step Computations

For each natural parameter update in the E-step, we need to compute $\Lambda = \mathrm{diag}(\mathbf{E}_q[\omega_i]) = \mathrm{diag}\big(\frac{\tanh(c_i/2)}{2c_i}\big)$ with

$$c_i = \sqrt{\mathbf{Var}_q[f(x_i)] + (\mathbf{E}_q[f(x_i)])^2}.$$

- $\mathbf{Var}_q[f(x_i)] = \Sigma_{ii}$. Use that if $E[z_i z_j] = \delta_{ij}$, then $E[z \odot (Az)] = diag(A)$.

- Applying $\alpha \mapsto (K^{-1} + \Delta) z$ and solving via CG+NUFFT is the recurring computational bottleneck.

2.6 Solving $z \mapsto \Sigma z$

Solve for α: **Version 1: $n \times n$ system**

$$\Sigma z = (K^{-1} + \Delta)^{-1} z = \alpha \iff (I + K\Delta)\,\alpha = Kz \iff (I + FD^2F^*\Delta)\,\alpha = FD^2F^*z.$$

Version 2: $m \times m$ system:

$$\begin{aligned}
(K^{-1} + \Delta)v = z &\iff (I + K\Delta)v = Kz && (9)\\
&\iff (I + \Phi\Phi^*\Delta)v = \Phi\Phi^*z && (10)\\
&\iff (I + \Phi\Phi^*\Delta)\Phi u = \Phi\Phi^*z && (11)\\
&\iff \Phi\left[u + \Phi^*\Delta\Phi u\right] = \Phi\Phi^*z && (12)\\
&\iff (I + \Phi^*\Delta\Phi)u = \Phi^*z \text{ solve for u} && (13)\\
&\text{solve the above for } u \text{ and substitute } v = \Phi u. && (14)
\end{aligned}$$

Note that in **??**, we can rearrange and see that $v \in \text{Range}(\Phi)$. Also in **??**, we can "cancel" the Φ on both sides since Φ is injective, since $\{\exp(i\omega_j x)\}_{j\in[M]}$ are linearly independent.

1. Each CG iteration requires two NUFFT applies of $I + DF^*\Delta FD$.
2. Plus two NUFFT setups.
3. We can speed up this solve by preconditioning with the approximation $\Delta = \delta I$ where δ is the mean of the Δ diagonal matrix. This cuts down computations dramatically if Δ is anywhere close to constant. Ie solve

   $$(I + \overline{\delta}\Phi^*\Phi)u = \Phi^*z$$

   In some arbitrary case I tested, from 100 CG solves down to 7.

3 M step

3.1 Hyperparameter Updates

From Wenzel Eq 5 ie Eq **??**:

$$\log p(y, \omega, f) = \left(\tfrac{1}{2}\tilde{y}^T f - \tfrac{1}{2} f^T \Omega f\right) + \log p(f) + \log p(\omega) + \text{const.}$$

After $\mathbf{E}_q[\cdot]$, only the prior term depends on θ:

$$\begin{aligned}
\frac{\partial}{\partial\theta}\mathcal{L} &= \frac{\partial}{\partial\theta}\mathbf{E}_q \log p(f) && (15)\\
&= \frac{\partial}{\partial\theta}\left(-\tfrac{1}{2}\,\operatorname{tr}\left((\Sigma + \mu\mu^T)K^{-1}\right) - \tfrac{1}{2}\log|K|\right) && (16)\\
&= \tfrac{1}{2}\,\mu^T K^{-1}(\partial_\theta K)\,K^{-1}\mu + \tfrac{1}{2}\,\operatorname{tr}\left(\Sigma\,K^{-1}(\partial_\theta K)\,K^{-1}\right) - \tfrac{1}{2}\,\operatorname{tr}\left(K^{-1}\partial_\theta K\right) && (17)\\
&= I + II + III. && (18)
\end{aligned}$$

3.2 Term 1

Let $v = K^{-1}\mu$. Then

$$\begin{aligned}
Kv = \mu \implies FD^2F^*\,v = \mu \implies F^*FD^2\,\beta = F^*\mu, \quad \text{where } \beta = F^*v, && (19)\\
\implies \text{solve Toeplitz } (F^*FD^2)\,\beta = F^*\mu, && (20)
\end{aligned}$$

So the first term becomes

$$\mu^T K^{-1}(\partial_\theta K)\,K^{-1}\mu = \left\langle \beta,\, D'\beta \right\rangle.$$

3.3 Term 2

$$II \approx \frac{1}{J}\sum_{j=1}^{J} z_j^T\,(K^{-1}+\Delta^{(t)})^{-1}K^{-1}FD'F^*\,K^{-1}z_j.$$

- Compute $F^*K^{-1}z$ as in Term 1.
- Then solve $z^T\Sigma$:
  - Solved the same system in E-step, but with $\Sigma_{(t-1)}$, would like to reuse this somehow

3.4 Term 3

$$III \approx \frac{1}{J}\sum_{j=1}^{J} z_j^T\,FD'F^*\,K^{-1}z_j,$$

Like term I, this reduces to calculating one NUFFT after computing $F^*K^{-1}z$.

3.5 Summary

- The bottleneck is applying $\Sigma v = (K^{-1}+\Delta^{(t)})v$, which costs 2 NUFFT.

4 Revised M Step

Use Woodbury inversion Lemma to get:

$$\Sigma = (K^{-1}+\Delta)^{-1} = K - K(I+\Delta K)^{-1}\Delta K.$$

Hence we have the identity:

$$K^{-1}\Sigma K^{-1} = K^{-1} - (I+\Delta K)^{-1}\Delta.$$

$$\begin{aligned}
\frac{\partial}{\partial\theta}\mathcal{L} &= \frac{\partial}{\partial\theta}\mathrm{E}_q \log p(f) && (21)\\
&= \frac{\partial}{\partial\theta}\Big(-\tfrac{1}{2}\,\mathrm{tr}\big((\Sigma+\mu\mu^T)K^{-1}\big) - \tfrac{1}{2}\log|K|\Big) && (22)\\
&= \tfrac{1}{2}\,\mu^T K^{-1}(\partial_\theta K)\,K^{-1}\mu + \tfrac{1}{2}\,\mathrm{tr}\big(\Sigma\,K^{-1}(\partial_\theta K)\,K^{-1}\big) - \tfrac{1}{2}\,\mathrm{tr}\big(K^{-1}\partial_\theta K\big) && (23)\\
&= I + II + III. && (24)
\end{aligned}$$

Cycling the traces:

$$\begin{aligned}
\tfrac{1}{2}\,\mathrm{tr}\big(\Sigma\,K^{-1}(\partial_\theta K)\,K^{-1}\big) &= \tfrac{1}{2}\,\mathrm{tr}\big(K^{-1}\Sigma\,K^{-1}(\partial_\theta K)\big) && (25)\\
&= \frac{1}{2}\,\mathrm{tr}(K^{-1}\partial_\theta K) - \frac{1}{2}\,\mathrm{tr}((I+\Delta K)^{-1}\Delta\partial_\theta K) && (26)
\end{aligned}$$

Hence we can cancel the third term and write:

$$\begin{aligned}
\frac{\partial}{\partial\theta}\mathcal{L} &= I - \frac{1}{2}\,\mathrm{tr}((I+\Delta K)^{-1}\Delta\partial_\theta K). && (27)\\
&= I - \frac{1}{2}\,\mathrm{tr}(\Delta FD'F^*(I+\Delta K)^{-1}) && (28)
\end{aligned}$$

This is an improvement over before since that inverse is well-conditioned. The idea to solve it efficiently is to approximate Δ as constant times identity. Let $\alpha := (I+\Delta K)^{-1}z$. Then $z = (I+\Delta FD^2)F^*\alpha$ and:

$$F^* z = (I + F^* \Delta F D^2) F^* \alpha \tag{29}$$
$$\approx (I + \overline{\delta} T D^2) F^* \alpha. \tag{30}$$

Solve the second system using Toeplitz machinery, then warm start the above system–should only take a few iterations to converge with this warm start.

Note actualy best to solve

$$(I + F^* \Omega F D^2) F^* \alpha = F^* z \tag{31}$$
$$\implies (I + D F^* \Omega F D) D F^* \alpha = D F^* z \tag{32}$$

so that we have a hermitian system and cg works better.

5 Predictive mean

As in Wenzel:

$$p(f_* | y) = \int p(f_* | f) p(f | y) df \tag{33}$$
$$\approx \int p(f_* | f) q(f) df \tag{34}$$
$$= N(f_* | \mu_*, \sigma_*^2), \tag{35}$$

where it is clearer to work in feature space. Let

$$\Psi := FD, \qquad K = \Psi \Psi^* = F D^2 F^*, \qquad \kappa := y - \tfrac{1}{2}, \qquad B := \Psi^* \Delta \Psi = D F^* \Delta F D.$$

Here:

$$F \in \mathbb{C}^{n \times M}$$

is the Fourier / NUFFT feature matrix at the training inputs,

$$D = \mathrm{diag}(ws) \in \mathbb{R}^{M \times M}$$

is the diagonal matrix of spectral weights, and

$$\Delta = \mathrm{diag}(\delta) \in \mathbb{R}^{n \times n}$$

is the diagonal PolyGamma variational parameter matrix. The Gaussian variational factor is

$$q(f) = N(\mu, \Sigma), \qquad \Sigma = (K^{-1} + \Delta)^{-1}, \qquad \mu = \Sigma \kappa.$$

Now define the feature-space mean coefficients by

$$m_u := (I + B)^{-1} \Psi^* \kappa \in \mathbb{C}^M,$$

and the candidate training-location mean

$$\tilde{\mu} := \Psi m_u.$$

We now show that $\tilde{\mu} = \mu$. First,

$$\mu = (K^{-1} + \Delta)^{-1} \kappa.$$

Using

$$(K^{-1} + \Delta)^{-1} = (I + K\Delta)^{-1} K,$$

we get

$$\mu = (I + K\Delta)^{-1} K\kappa.$$

Next, since $K = \Psi\Psi^*$ and

$$(I + AB)^{-1} A = A(I + BA)^{-1},$$

with

$$A = \Psi, \qquad B = \Psi^*\Delta,$$

we obtain

$$(I + K\Delta)^{-1}\Psi = (I + \Psi\Psi^*\Delta)^{-1}\Psi = \Psi(I + \Psi^*\Delta\Psi)^{-1}.$$

Therefore

$$(I + K\Delta)^{-1}K = (I + K\Delta)^{-1}\Psi\Psi^* = \Psi(I + \Psi^*\Delta\Psi)^{-1}\Psi^* = \Psi(I + B)^{-1}\Psi^*.$$

Substituting this into the expression for μ gives

$$\mu = \Psi(I + B)^{-1}\Psi^*\kappa = \Psi m_u = \tilde{\mu}.$$

Hence the posterior mean at the training locations is indeed

$$\mu = \Psi m_u.$$

Equivalently, the posterior mean satisfies

$$\left(K^{-1} + \Delta\right)\mu = \kappa, \qquad K^{-1}\mu = \kappa - \Delta\mu.$$

Applying Ψ^* gives

$$\Psi^* K^{-1}\mu = \Psi^*(\kappa - \Delta\mu).$$

Since $\mu = \Psi m_u$, this becomes

$$\Psi^* K^{-1}\mu = \Psi^*\kappa - \Psi^*\Delta\Psi m_u.$$

But from

$$(I + B)m_u = \Psi^*\kappa,$$

we also have

$$m_u = \Psi^*\kappa - \Psi^*\Delta\Psi m_u.$$

Comparing the last two displays shows that

$$m_u = \Psi^* K^{-1}\mu.$$

Multiplying by D^{-1} and using $\Psi = FD$ yields the equivalent relation

$$\beta^{(1)} := F^* K^{-1}\mu, \qquad m_u = D\beta^{(1)}.$$

For a new point x_*, define the unweighted Fourier row

$$(\varphi_*)_j := \exp(2\pi i\langle \xi_j, x_*\rangle), \qquad j \in [M],$$

and the weighted feature vector

$$(\psi_*)_j := \sqrt{h^d \hat{k}(\xi_j)}\, \exp(2\pi i\langle \xi_j, x_*\rangle), \qquad j \in [M].$$

Equivalently,

$$\psi_* = D\varphi_*.$$

The cross-covariance vector between the training inputs and x_* is

$$k_* := K(X, x_*) = \Psi\psi_* = FD^2\varphi_*.$$

Then the predictive latent mean is

$$\mu_* = \mathbf{E}[f_*|y] \approx k_*^T K^{-1}\mu.$$

Using the identities above,

$$\begin{aligned} \mu_* &= k_*^T K^{-1}\mu && (36)\\ &= \psi_*^* \Psi^* K^{-1}\mu && (37)\\ &= \psi_*^* m_u. && (38) \end{aligned}$$

Hence

$$\boxed{\mu_* = \psi_*^* m_u.}$$

To connect back to Term 1 in the M-step, define

$$\beta^{(1)} := F^* K^{-1}\mu,$$

Therefore

$$\mu_* = \varphi_*^* D m_u = \varphi_*^* D^2 \beta^{(1)}.$$

In practice, for a batch of test locations this is a single Type-2 NUFFT.

6 Predictive variance

Starting from the conditional GP variance formula and integrating against $q(f) = N(\mu, \Sigma)$ gives

$$\sigma_*^2 = k_{**} - k_*^T K^{-1} k_* + k_*^T K^{-1}\Sigma K^{-1} k_*.$$

Using the revised M-step identity

$$K^{-1}\Sigma K^{-1} = K^{-1} - (I + \Delta K)^{-1}\Delta,$$

the K^{-1} terms cancel, so

$$\boxed{\sigma_*^2 = k_{**} - k_*^T (I + \Delta K)^{-1}\Delta k_*.}$$

Thus we never need to store or apply K^{-1} explicitly.

In the feature notation above, let

$$S_u := (I + B)^{-1}.$$

Since $\Sigma = \Psi S_u \Psi^*$, the predictive variance can be written as

$$\boxed{\sigma_*^2 = \psi_*^* S_u \psi_* = \psi_*^* (I + B)^{-1} \psi_*.}$$

So predictive variances reuse the same Hermitian operator $I + DF^*\Delta FD$ as the revised M-step: for each test point, solve

$$(I + B)u_* = \psi_*,$$

then compute

$$\sigma_*^2 = \psi_*^* u_*.$$

For a batch of test points, solve these systems simultaneously with batched CG.

7 Predictive Class Probability

Finally, the predictive class probability is

$$p(y_* = 1|y) = \int \sigma(f_*)\, p(f_*|y)\, df_* \approx \int \sigma(f_*)\, N(f_*|\mu_*, \sigma_*^2)\, df_*,$$

which is a one-dimensional integral. One may evaluate it by quadrature, or use the same approximation as at the training points:

$$p(y_* = 1|y) \approx \sigma\left(\frac{\mu_*}{\sqrt{1 + \pi\sigma_*^2/8}}\right).$$

8 Summary

We can learn hyperparameters, compute predictive means in $\mathcal{O}(n + M \log M)$, but the cost of applying $K^{-1} + \Delta$ will give a larger constant than in the conjugate case, since we must rely on NUFFT in conjugate gradient solves, rather than the Toeplitz operation.

9 Negative Binomial

We can use the same PolyGamma augmentation for negative binomial observations. The key point is that the Fourier / NUFFT linear algebra is unchanged; only the likelihood-dependent scalars in the E-step change, and if we choose to learn the shape parameter r, that is an additional scalar update in the M-step.

9.1 Likelihood and augmentation

Let

$$y_i \mid f_i, r \sim \mathrm{NB}(r, \sigma(f_i)),$$

with pmf

$$p(y_i \mid f_i, r) = \binom{y_i + r - 1}{y_i} \sigma(f_i)^{y_i} \left(1 - \sigma(f_i)\right)^r.$$

Using

$$\sigma(f) = \frac{e^f}{1 + e^f}, \qquad 1 - \sigma(f) = \frac{1}{1 + e^f},$$

the f_i-dependent part is

$$p(y_i \mid f_i, r) \propto \frac{e^{y_i f_i}}{(1 + e^{f_i})^{y_i + r}}.$$

Apply the PolyGamma identity with

$$a = y_i, \qquad b = y_i + r, \qquad \kappa_i := a - \frac{b}{2} = \frac{y_i - r}{2}.$$

Then

$$\frac{e^{y_i f_i}}{(1 + e^{f_i})^{y_i + r}} = 2^{-(y_i + r)} e^{\kappa_i f_i} \int_0^\infty \exp\!\left(-\frac{1}{2}\omega_i f_i^2\right) PG(\omega_i \mid y_i + r, 0)\, d\omega_i.$$

Hence the augmented joint is

$$p(f, y, \omega \mid r, \theta) = p(f \mid \theta) \prod_{i=1}^n \binom{y_i + r - 1}{y_i} 2^{-(y_i + r)}\, e^{\kappa_i f_i}\, e^{-\frac{1}{2}\omega_i f_i^2}\, PG(\omega_i \mid y_i + r, 0).$$

9.2 Mean-field updates

Use the same variational family

$$q(f, \omega) = q(f) \prod_{i=1}^n q(\omega_i).$$

Conditioning on $q(\omega)$ gives

$$\log q^*(f) = -\frac{1}{2} f^T K_\theta^{-1} f + \kappa^T f - \frac{1}{2} f^T \Lambda f + \text{const},$$

where

$$\kappa := \left(\frac{y_1 - r}{2}, \dots, \frac{y_n - r}{2}\right)^T, \qquad \Lambda := \mathrm{diag}\big(\mathbf{E}_q[\omega_i]\big).$$

Therefore

$$q(f) = N(\mu, \Sigma), \qquad \Sigma = (K_\theta^{-1} + \Lambda)^{-1}, \qquad \mu = \Sigma\kappa.$$

Fixing $q(f)$ and using the same tilting identity as in the Bernoulli case,

$$q^*(\omega_i) = PG(y_i + r, c_i),$$

where

$$c_i := \sqrt{\mathbf{E}_q[f_i^2]} = \sqrt{\mu_i^2 + \Sigma_{ii}}.$$

Its mean is

$$\mathbf{E}_q[\omega_i] = \frac{y_i + r}{2c_i} \tanh\left(\frac{c_i}{2}\right).$$

Thus the only changes relative to Bernoulli are

$$\kappa_i = \frac{y_i - r}{2}, \qquad b_i = y_i + r, \qquad \Lambda_{ii} = \frac{y_i + r}{2c_i} \tanh\left(\frac{c_i}{2}\right).$$

All of the Fourier-feature machinery from the Bernoulli model is unchanged:

1. $\Sigma z = (K^{-1} + \Lambda)^{-1} z$ is still applied with the same $M \times M$ Hermitian solve,
2. $\mathrm{diag}(\Sigma)$ is still estimated by Hutchinson probes,
3. the Toeplitz warm start still uses the approximation $\Lambda \approx \overline{\lambda} I$.

9.3 Kernel hyperparameter update

For fixed $q(f)$, the kernel hyperparameter gradient is exactly the same as in the Bernoulli case, because only the prior term depends on θ:

$$\frac{\partial}{\partial \theta} \mathcal{L} = \frac{\partial}{\partial \theta} \mathbf{E}_q[\log p(f \mid \theta)].$$

Therefore the revised M-step formula from the previous section applies without modification:

$$\frac{\partial}{\partial \theta} \mathcal{L} = \frac{1}{2} \mu^T K^{-1} (\partial_\theta K) K^{-1} \mu - \frac{1}{2} \mathrm{Tr}\left((I + \Lambda K)^{-1} \Lambda \, \partial_\theta K\right).$$

So the same Term 1 / trace-estimation machinery, NUFFTs, and warm-started CG are reused.

9.4 Updating the shape parameter r

Now suppose r is also learned. For fixed $q(f)$, the only part of the ELBO that depends on r is the expected log-likelihood:

$$\mathcal{L}(r) = \sum_{i=1}^{n} \mathbf{E}_{q(f_i)} \left[\log p(y_i \mid f_i, r)\right] + \text{const}.$$

Using

$$\log p(y_i \mid f_i, r) = \log \Gamma(y_i + r) - \log \Gamma(r) - \log(y_i!) + y_i f_i - (y_i + r) \log(1 + e^{f_i}),$$

we differentiate with respect to r while holding $q(f)$ fixed:

$$\frac{\partial \mathcal{L}}{\partial r} = \sum_{i=1}^{n} \left[\psi(y_i + r) - \psi(r) - \mathbf{E}_{q(f_i)} \log(1 + e^{f_i})\right].$$

Since

$$\log \sigma(-f) = -\log(1 + e^f),$$

this is equivalently

$$\boxed{\frac{\partial \mathcal{L}}{\partial r} = \sum_{i=1}^{n} \left[\psi(y_i + r) - \psi(r) + \mathbf{E}_{q(f_i)} \log \sigma(-f_i)\right].}$$

Under the Gaussian variational approximation

$$q(f_i) = N(\mu_i, \Sigma_{ii}),$$

the remaining expectation is a one-dimensional Gaussian integral:

$$\mathbf{E}_{q(f_i)} \log \sigma(-f_i) = \int \log \sigma(-f)\, N(f \mid \mu_i, \Sigma_{ii})\, df.$$

Writing $f = \mu_i + \sqrt{\Sigma_{ii}}\, t$ with $t \sim N(0,1)$ gives

$$\mathbf{E}_{q(f_i)} \log \sigma(-f_i) = \int \log \sigma\big(-(\mu_i + \sqrt{\Sigma_{ii}}\, t)\big)\, N(t \mid 0,1)\, dt,$$

which is evaluated numerically by Gauss–Hermite quadrature. This is only $\mathcal{O}(nK)$ for K quadrature nodes and requires no NUFFT, Toeplitz, or CG solve once μ_i and Σ_{ii} are already available from the E-step.

To enforce positivity, we optimize

$$\eta := \log r.$$

By the chain rule,

$$\frac{\partial \mathcal{L}}{\partial \eta} = \frac{\partial \mathcal{L}}{\partial r}\frac{\partial r}{\partial \eta} = r\,\frac{\partial \mathcal{L}}{\partial r}.$$

Hence a gradient-ascent update takes the form

$$\eta^{(t+1)} = \eta^{(t)} + \rho_r\, r^{(t)} \sum_{i=1}^{n} \left[\psi(y_i + r^{(t)}) - \psi(r^{(t)}) + \mathbf{E}_{q(f_i)} \log \sigma(-f_i)\right].$$

After updating r, the next E-step uses the refreshed quantities

$$\kappa_i = \frac{y_i - r}{2}, \qquad b_i = y_i + r, \qquad \Lambda_{ii} = \frac{y_i + r}{2c_i} \tanh\!\left(\frac{c_i}{2}\right).$$

So learning r changes the next diagonal system through κ and Λ, but the expensive linear algebra is exactly the same as before.

9.5 Predictive mean count

In this parameterization,

$$\mathbf{E}[y_i \mid f_i, r] = r e^{f_i}.$$

Therefore, under the Gaussian approximation

$$f_* \mid y \approx N(\mu_*, \sigma_*^2),$$

the predictive mean count is

$$\mathbf{E}[y_* \mid y] \approx r\,\mathbf{E}[e^{f_*}] = \boxed{r \exp\left(\mu_* + \frac{1}{2}\sigma_*^2\right).}$$

Thus prediction reuses the same latent predictive mean μ_* and variance σ_*^2 derived above; only the final response map differs from the Bernoulli case.